Stat Model A
F79MA 2025-26 - Assessed Project 1

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1. Introduction

This report discusses whether the maximum likelihood estimator (MLE) $\hat{\theta}$ is appropriate for use in practice for financial risk modeling. Utilising both theoretical analysis and simulations in R, this report shall focus on the Laplace distribution centered at θ, with the scale parameter set as 1.

2. Mathematical Derivations

The distribution under consideration is defined by the probability density function (pdf):

$$f(x;\theta) = \frac{1}{2}e^{-|x-\theta|}, \quad -\infty < x < \infty, \quad \theta \in \mathbb{R}$$

and the cumulative distribution function (cdf)

$$F(x;\theta) = \begin{cases} \frac{1}{2}e^{(x-\theta)} & x < \theta \\ 1 - \frac{1}{2}e^{(-(x-\theta))} & x > \theta \end{cases}$$

where θ is the location parameter of the distribution.

2.1 The likelihood and log-likelihood

Assuming henceforth that the sample $\underline{x} = (x_1, \ldots, x_n)$ is independent and identically distributed, and assuming that the density function $f(x;\theta)$ and its partial derivative $\frac{\partial}{\partial\theta}f(x;\theta)$ are continuous in $\underline{x}$ and θ, the likelihood function is given by:

$$\begin{aligned} L(\theta;\underline{x}) &= \prod_{i=1}^{n} \frac{1}{2}e^{-|x_i-\theta|} \\ &= \left(\frac{1}{2}\right)^n e^{-\sum_{i=1}^{n}|x_i-\theta|} \end{aligned}$$

And the log-likelihood function is:

$$\begin{aligned} l(\theta;\underline{x}) &= \ln(L(\theta;\underline{x})) \\ &= -n\ln 2 - \sum_{i=1}^{n}|x_i-\theta| \end{aligned}$$

2.2 The score function, Fisher information and Cramér–Rao Lower Bound

For sample $\underline{x} = (x_1, \ldots, x_n)$, the score function is given by:

$$U(\theta) = \frac{d}{d\theta}l(\theta;\underline{x})$$

Let $u = x_i - \theta$ for $i = 1, \ldots, n$, then:

$$\begin{aligned} U(\theta) &= \sum_{i=1}^{n} -\frac{d|u|}{du} \cdot \frac{du}{d\theta} \\ &= \sum_{i=1}^{n} \begin{cases} -1 \cdot 1 \cdot -1 & u > 0 \\ -1 \cdot -1 \cdot -1 & u < 0 \end{cases} \\ &= \sum_{i=1}^{n} \begin{cases} 1 & x_i > \theta \\ -1 & x_i < \theta \end{cases} \end{aligned}$$

The Fisher information therefore follows by:

$$\begin{aligned} I(\theta) &= \mathbb{E}[U(\theta)^2] \\ &= \sum_{i=1}^{n} \mathbb{E}\left[\begin{cases} 1 & x_i > \theta \\ 1 & x_i < \theta \end{cases} \right] \\ &= \sum_{i=1}^{n} \mathbb{E}[1] \quad \forall x_i \neq \theta \\ &= n \quad \forall x_i \neq \theta \end{aligned}$$

By letting $T(\underline{x})$ be an unbiased estimator of θ, the Cramér–Rao Lower Bound (CRLB) is $\text{Var}(T) \geq \frac{1}{n} \quad \forall x_i \neq \theta$.

2.3 The Maximum Likelihood Estimator

Based on the sample $\underline{x} = (x_1, \ldots, x_n)$, we can find the maximum likelihood estimator (MLE) $\hat{\theta}$ by letting $U(\theta) = 0$. Since the Score function is a summation of 1 and -1s depending on whether the sample is strictly more than or strictly less than theta, we will want to choose θ as the point that divides the sample into 2 halves. This will therefore generate 2 cases.

Case 1, when n is even, then $\hat{\theta} \in \left[x_{(\frac{n}{2})}, x_{(\frac{n}{2}+1)} \right]$

Case 2, when n is odd, then $\hat{\theta} = x_{(\frac{n+1}{2})}$

Thus, showing that the MLE corresponds to the sample median.

The CRLB is not attainable in this case because $U(\theta; \underline{x})$ cannot be expressed in the form $U(\theta; \underline{x}) = a(\theta)[T(\theta) - \theta]$ for some function $a(\theta)$, with $T(\theta)$ being an unbiased estimator of θ.

3. Simulations in R

Seed value currently in use is 6014. To conduct the simulations, pseudorandom data of 1000 simulations each with 100 sample sizes were generated using inverse transform method. Summary of results obtained are as follows:

- `Quantity1` = 0.0572412
- `Quantity2` = -201.508

- `Quantity3` = -0.107244
- `Quantity4` = 0.221727
- `Quantity5` = -0.1546
- `Quantity6` = 0.218
- `Quantity7` = -0.000862685
- `Quantity8` = 0.0117776
- `Quantity9` = 0.0117666
- `Quantity10` = 0.01

The report henceforth will explain the methodology used to obtain these figures, while also providing relevant comments and demonstrations where necessary.

3.1 The maximum likelihood estimate $\hat{\theta}$ for the first simulated sample (`Quantity1`)

The first sample is first segregated and renamed as `first_sample`.

As per findings in **2.3**, the MLE corresponds to the sample median. Thus, median function built into R was directly called to compute the MLE for the first simulated sample. Therefore obtaining `Quantity1`.

3.2 The log-likelihood for `Quantity1` (`Quantity2`)

`Quantity2` is directly obtained from substituting `Quantity1` into the log-likelihood formula as per findings in **2.1**.

In order to demonstrate that `Quantity1` maximises `Quantity2`, we first generate a range of possible values of theta bounded between 1 less of the minimum and 1 more of the maximum of the first sample. These values are stored in `theta_grid`, to be used to plot the x-axis of the graph.

Then, a log-likelihood function is defined as `log_likelihood` with parameters `first_sample_count, first_sample` and `theta` for future ease of use.

A for loop is then used to find the corresponding values if each value of `theta_grid` were plugged into the `log_likelihood` function, to be used to plot the y-axis of the graph.

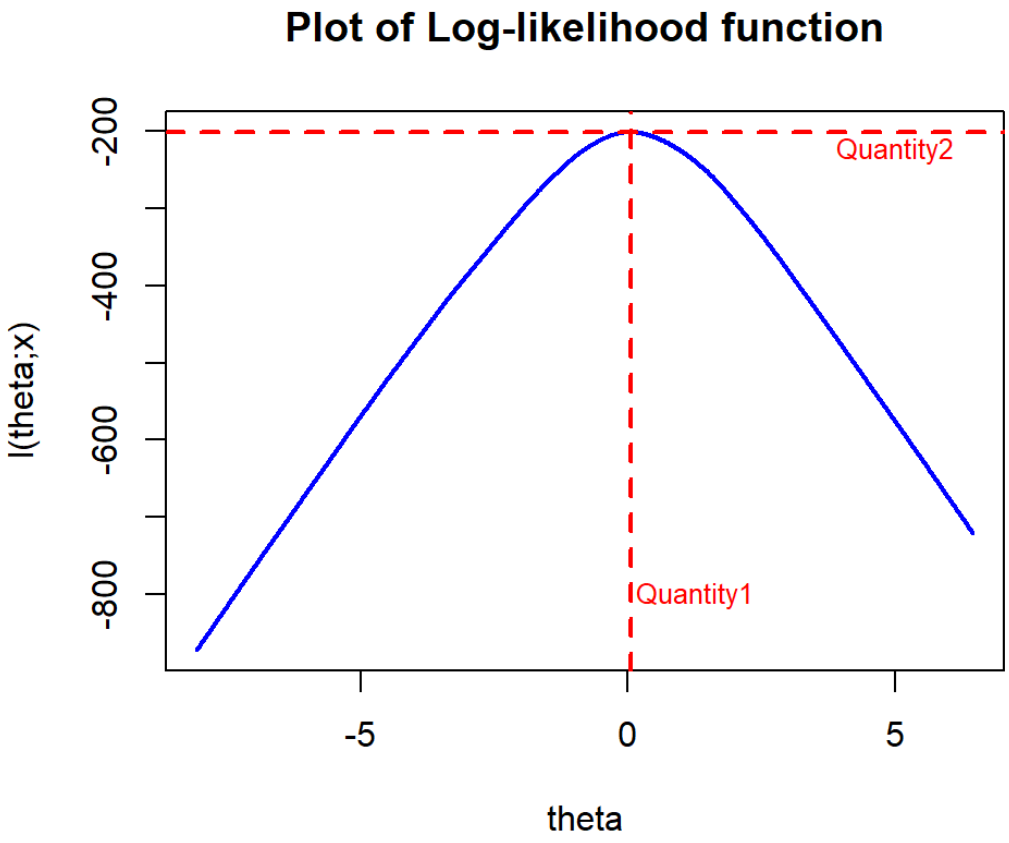


Figure 1: log-likelihood graph

The log-likelihood graph with a range of theta is then plotted (Figure 1) while showing the intersection of values `Quantity1` and `Quantity2`. This is to demonstrate that `Quantity1` maximises log-likelihood as these 2 quantities intersect at the maximum vertex of the graph (Figure 1).

3.3 90% Confidence Interval for θ using the asymptotic normality of the MLE (`Quantity3, Quantity4`)

By the asymptotic normality of the MLE, the following result is true:

$$\frac{\hat{\theta} - \theta}{\sqrt{I(\theta)^{-1}}} \to N(0,1), \quad \text{as } n \to \infty$$

By defining the z-score appropriately, while deriving the Fisher information as shown in **2.2** as variable `I`, the Lower and Upper limit of the 90% Confidence Interval can be derived using the formula below:

$$\hat{\theta} \pm z_{\frac{\alpha}{2}} \sqrt{I(\hat{\theta})^{-1}}$$

Hence obtaining `Quantity3` and `Quantity4`.

3.4 90% Confidence Interval for θ using asymptotic distribution of the likelihood-ratio (deviance) statistic (`Quantity5, Quantity6`)

By the asymptotic distribution of the deviance statistic, the following holds true:

$$D(\theta) = 2[I(\hat{\theta} - I(\theta)] \to \chi_1^2 \quad \text{as } n \to \infty$$

Thus, an approximate large-sample $100(1-\alpha)\%$ confidence interval for θ is as such:

$$\{\theta : D(\theta) < \chi_1^2(\alpha)\}$$

Therefore, the threshold $\chi_1^2(0.1)$ named `thresh` was defined. Then, using a range of possible θ values, multiple $D(\theta)$ named as `dev` were calculated and all values lower than `thresh` were stored in another variable `ci`.

The lowest and highest of the range of values in `ci` then form the Lower and Upper Limits of the 90% Confidence Interval, obtaining `Quantity5` and `Quantity6`.

Notice that the confidence interval obtained from the asymptotic normality approach is much narrower, compared to the deviance approach. The deviance approach however, arguably generates to lowest Upper Limit, at least in this instance. Therefore, utillising both approaches to conclude a final 90% confidence interval of $(-0.107244, 0.218)$ enables the most precise population estimates.

3.5 The bias (`Quantity7`), variance (`Quantity8`), and mean squared error (MSE) (`Quantity9`) of the MLE estimates

Now, falling back, using the entire sample matrix with 1000 simulations of 100 sample spaces, the `theta_hat_vector` is first calculated by looping through each simulation calculating the MLE for each instance.

Thereafter, the bias, variance and MSE was then calculated using the following formulas:

$$Bias(\hat{\theta}) = \mathbb{E}[\hat{\theta} - \theta]$$
$$MSE(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2]$$

With variance being calculated with the built-in `Var()` function in R. Hence, obtaining `Quantity7` , `Quantity8` and `Quantity9`.

3.6 The CRLB for a smooth unbiased estimator for a sample of size n (`Quantity10`)

As per findings in **2.2**, it has been shown that the CRLB for a smooth unbiased estimator for the sample of size n would be $\frac{1}{n}$. Hence, this formula was used with knowledge of $n = 100$ to directly obtain `Quantity10`.

3.7 Plots on Bias, Variance, and MSE of the MLE as n increases

First, a general function to calculate the `theta_hat_vector_new` is created.

Then, the given code is put inside a loop repeating 1000 times, modified only by adding 1 to the previous sample size of each of the 1000 simulations. This results in sample

matrices with a range of 100 to 1100 sample sizes, all with 1000 simulations. With every iteration, the respective Bias, Variance and MSE for each iteration is calculated and stored as a vector.

Next, plots on Bias, Variance and MSE of the MLE as n increases are plotted as shown in Figures 2, 3 and 4 below respectively.

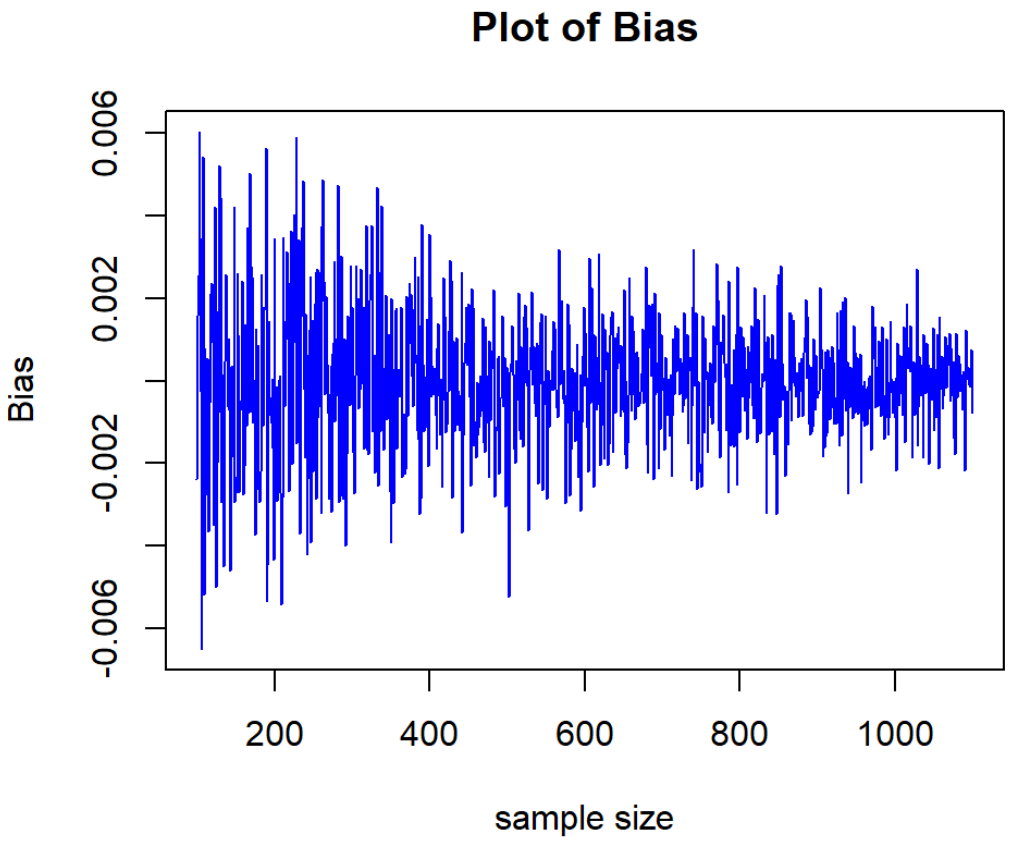


Figure 2: Bias against sample size n

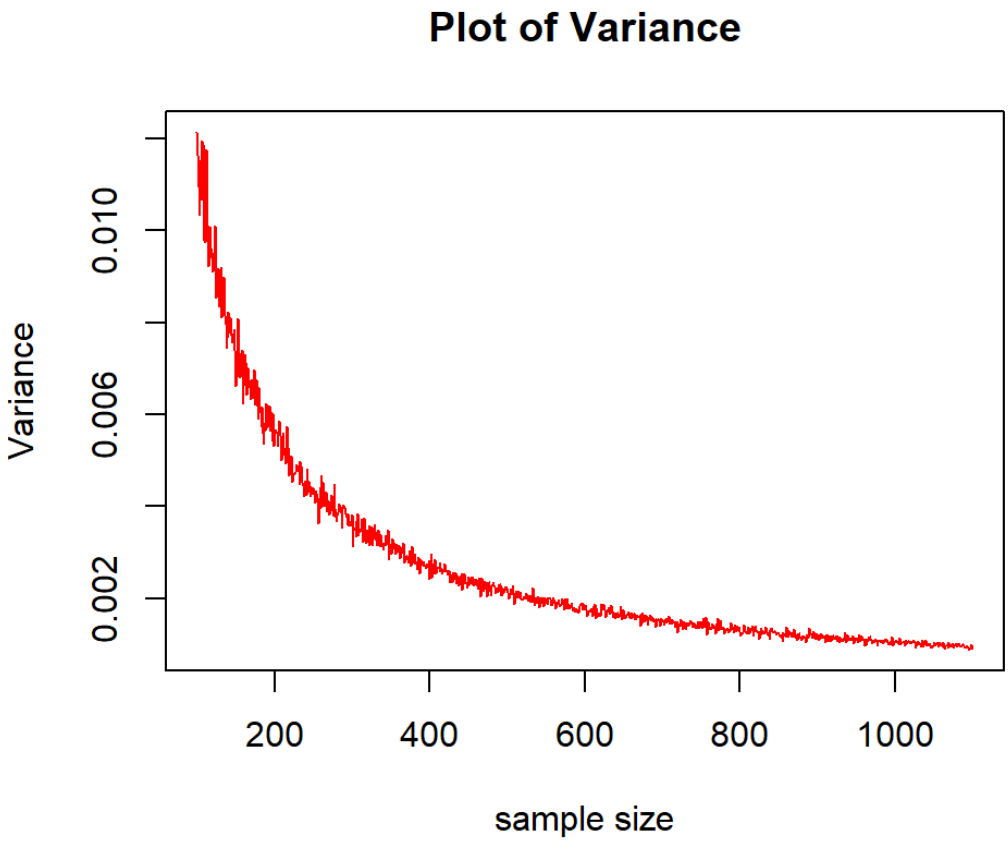


Figure 3: Variance against sample size n

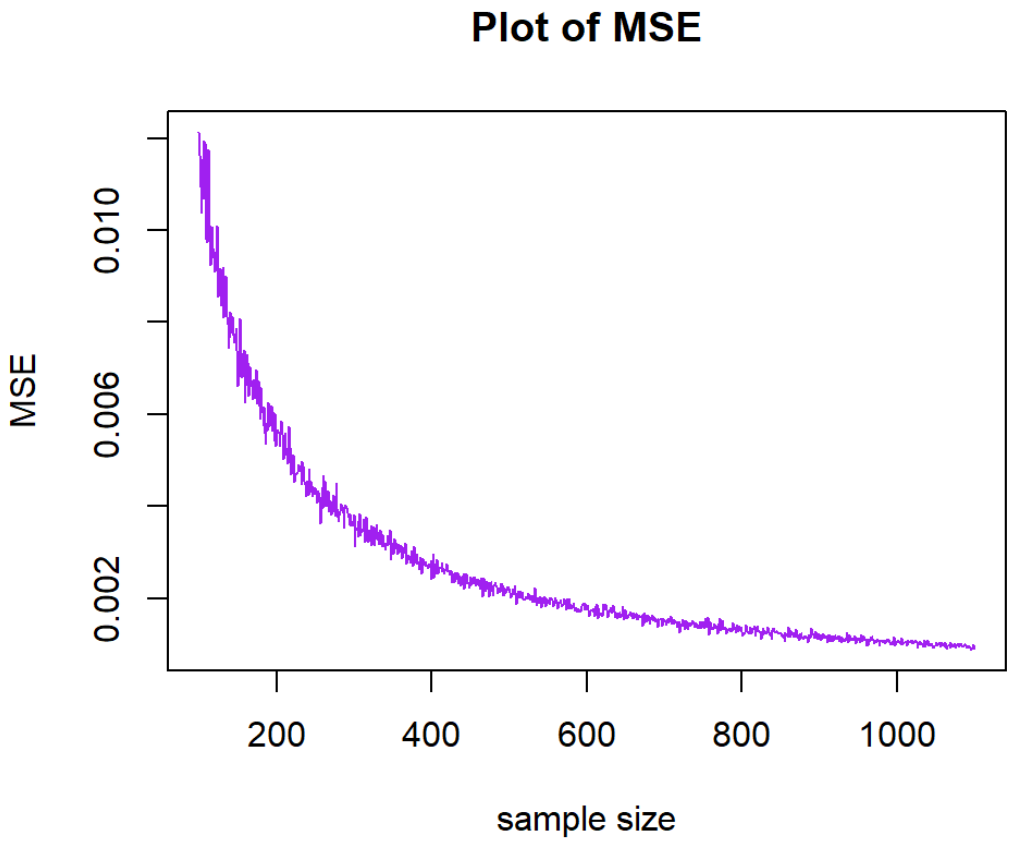


Figure 4: MSE against sample size n

From Figures 2, 3 and 4, it can be concluded that the MLE appears consistent and efficient. This is because as the sample size increases, the bias, variance and MSE seem to approach 0.

4. Conclusion

From this exploration, it has been shown that the MLE, a method to obtain values for parameters of a probability distribution, is very appropriate for use in practice. This is because its properties provide a solution to enable extremely precise population estimates to be generated. Furthermore, as the sample size increases, the MLE shows properties of consistency and efficiency. This result will prove useful in modeling certain heavy-tailed financial risks, where sample sizes are abundant.